APPZYS TECHNOLOGIES PRIVATE LIMITED

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PROJECT SPECIFICATION & MASTER ENGINEERING BLUEPRINT

APPZYS MOBILE APP DEVELOPMENT & AI/OCR STUDIO WEB PLATFORM

Complete Functional Specification, Competitive Gap Analysis, Full-Stack Architecture, Database Schema, REST API Contract, Security Plan, SEO Strategy & 18-Day Execution Roadmap

DOCUMENT FIELD	SPECIFICATION VALUE
Document Reference	`APPZYS-SRS-2026-V2.0` (Formal Monochrome Specification)
Project Objective	Transform `appzys.com` from a consumer utility showcase into an enterprise B2B Mobile Application & AI/OCR Development Agency platform.
Technology Stack	React 19 + Vite 6 + Tailwind CSS v4 | Java 21 + Spring Boot 3.4.3 | MySQL / TiDB Cloud | Docker + Render
Implementation Status	**Phase 1 (Days 1–2): Completed & Build-Verified | Phase 2 (Days 3–18): Planned Roadmap**
Public Positioning Rule	100% Mobile App Engineering focus (iOS, Android, Flutter, React Native, AI/OCR, ASO); Senior-led studio (no headcount disclosure).

EXECUTIVE ABSTRACT

Appzys Technologies Private Limited has published consumer mobile applications on Google Play and the Apple App Store under the developer identity *"Appzys - AI Apps, Scanner & OCR"*, achieving approximately 900K–1M+ combined installs led by *Copy Text On Screen* (~700K+ installs, ~4.1★ average rating). While this store footprint establishes production mobile engineering credibility, the existing web presence lacks structured B2B client acquisition workflows, interactive estimation tools, verified case-study pages, technical SEO schema, and backend lead persistence. This document defines the complete engineering plan, exact page-by-page website behavior, benchmark gap closures against top-10 mobile agencies, implemented Days 1–2 modules, and the Days 3–18 delivery roadmap.

FORMAL DOCUMENT APPROVAL & SIGN-OFF

Organization: Appzys Technologies Pvt. Ltd.	**Architecture Baseline:** Days 1–2 Verified Build	**Classification:** Internal Project Blueprint

FRONT MATTER

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HOW TO READ STATUS MARKERS IN THIS SPECIFICATION

Every functional module, database table, and REST endpoint in this report is marked with one of two explicit engineering verification tags:

- **BUILT** / **COMPLETED** — Implemented in codebase during Days 1–2 and verified via `npm run build` (Vite 6) and `mvn compile` (Java 21).
- **PLANNED** — Scoped for Sprint 2 (Days 3–18) covering SEO depth, Admin JWT portal, Blog CMS, automated E2E tests, and final DNS cutover.

CHAPTER 1

PROBLEM STATEMENT, EXISTING ASSETS & BUSINESS SCOPE

1.1 Organization & Existing Product Baseline

Appzys Technologies Private Limited (incorporated circa December 2021, registered under ROC Chennai, Tamil Nadu, India) operates a live portfolio of consumer mobile utilities across Google Play and the Apple App Store. Unlike early-stage software agencies that rely solely on mock UI concepts, Appzys already maintains production software used by nearly one million end users.

ASSET CATEGORY	OBSERVED PRODUCTION ASSET	STRATEGIC ROLE IN NEW B2B PLATFORM
Developer Store Profile	`Appzys - AI Apps, Scanner & OCR` (Google Play & Apple App Store)	Verified E-E-A-T trust proof (~900K–1M+ combined installs) linked directly from B2B case studies.
Flagship Mobile App	*Copy Text On Screen* (~700K+ installs, ~4.1★ from 1,500+ reviews, 100+ languages)	Powers Hero Phone Interactive Demo (OCR Screen Scan + 100+ Language AI) and primary Case Study #1.
Additional AI/OCR Suite	*AI Text Scanner: Image To Text*, *Handwriting to Text Scanner*, PDF Document Scanners, AI Chat	Case Studies #2 and #3 proving multi-format computer vision, on-device latency optimization, and store retention.
Legacy Legal Routes	`/ios/textscanner/privacypolicy`, `/ios/textscanner/termsofuse`, `/ios/handocr/*`, `/ios/aichat/*`, `/ios/copytext/privacypolicy`	**Zero-Downtime Preservation Rule:** All existing App Store legal URLs must remain active so store listings are never broken.

1.2 Core Problem Statement

Despite strong consumer app metrics, the legacy `appzys.com` website behaves strictly as a static app-policy host rather than a B2B revenue funnel. Prospective founders and enterprise buyers visiting the domain cannot inspect service offerings, test live Appzys AI/OCR capabilities, estimate app budgets, read architectural case studies, or submit qualified project inquiries stored in a central database.

CURRENT PROBLEMS (BEFORE PROJECT)

- No structured B2B landing page, mobile services catalog, or conversion CTA flow.
- No interactive proof of Appzys's real OCR / AI engineering latency or multi-language support.
- Zero budget or timeline self-estimation tools; leads drop off when pricing is opaque.
- No database persistence, rate limiting, spam protection, or automated email alerts for inquiries.
- No structured JSON-LD schema, XML sitemap, or targeted SEO keyword landing architecture.

TARGET SOLUTIONS (THIS PROJECT)

- **Proof-First B2B Studio Website:** Interactive Hero Phone Demo + 700K+ / 1M+ store proof.
- **Self-Serve Cost Estimator:** Transparent 4-step budget & week calculator persisted in MySQL.
- **Database-Linked Lead Flow:** Seamless link between Cost Estimate ID and Contact Submission.
- **Full-Stack Production Architecture:** React/Vite SPA + Java Spring Boot REST API + TiDB Cloud.
- **Strict Mobile Positioning:** Exclusively presents mobile & AI/OCR app engineering services to clients.

1.3 Scope Boundaries & Positioning Rules

MANDATORY PROJECT CONSTRAINTS

1. **Strict Mobile-Only Public Scope:** All public service cards, hero copy, and Tech Stack Tabs must showcase *Mobile App Engineering* (Native iOS/Android, Flutter, React Native, AI/OCR Vision, Mobile UI/UX, App Store & ASO). Generic backend/cloud outsourcing tabs were explicitly removed from the public Tech Stack UI.
2. **Zero Headcount Disclosure:** Public pages must present Appzys as a senior-led Mobile & AI App Studio. Internal team counts (such as "5-member") are permanently excluded from public UI copy.
3. **Full-Stack Internal Implementation:** Even though the company sells mobile applications externally, the company website itself is engineered as a secure full-stack system (React + Tailwind + Java 21 Spring Boot + MySQL/TiDB Cloud + Docker + Render).

1.4 What Our Company Can Deliver (Studio Capability Matrix)

SERVICE LINE	MOBILE ENGINEERING DELIVERABLES	TARGET ENGAGEMENT MODEL
Native iOS & Android Apps	Swift, SwiftUI, Kotlin, Jetpack Compose, offline-first SQLite/Room, smooth 60fps UI	Fixed-Scope MVP (4–8 Weeks) or Dedicated Senior Pod
Cross-Platform Mobile	Flutter & React Native single-codebase iOS + Android applications with native bridges	Fixed-Scope MVP Sprint (Startup Speed)
AI, OCR & Computer Vision	On-device ML Kit / TensorFlow Lite / CoreML, 100+ language OCR, handwriting recognition, LLM chat	AI & OCR Modernization Sprint (2–5 Weeks)
Mobile UI/UX & ASO Growth	Interactive Figma prototypes, design systems, Play Store / App Store review compliance, ASO launch	End-to-End Discovery to Store Scale

CHAPTER 2

REFERRAL & TOP-10 COMPETITOR ANALYSIS (GAP CLOSURE MATRIX)

2.1 Benchmark Websites Studied

To match top-ranking mobile application agencies on search engines and conversion rates, two primary referral websites and leading global/Indian mobile development agencies were audited during Phase 1 research:

- **Primary Referral Benchmarks:** `orangemantra.com` (enterprise trust bars, domain-focused service cards, sticky conversion CTAs) and `themaestro.in` (clean studio visual hierarchy, agile sprint clarity, direct founder contact pathways).
- **Top-10 Organic Competitor Set:** `appinventiv.com`, `hyperlinkinfosystem.com`, `konstantinfo.com`, `techaheadcorp.com`, `mindinventory.com`, `simform.com`, and `theninehertz.com`.

2.2 Features Adopted From Competitors vs. Gaps Appzys Solves

Large multi-thousand-employee agencies dominate generic keywords via heavy content volume, yet they suffer from slow, template-heavy sales funnels, hidden pricing, and stock mockups instead of owned consumer products. The matrix below documents exactly what Appzys adopts from top performers and what unique differentiators Appzys builds to outperform them.

EVALUATION DIMENSION	WHAT TOP-10 COMPETITORS & REFERRALS DO	WHAT IS MISSING THERE & WHAT APPZYS IMPLEMENTS
Hero Section Proof	Static lifestyle banners or generic 3D illustrations with vague slogans (“We Build Future Apps”).	**Appzys Innovation:** Live interactive *3-Tab Smartphone Showcase* right in the Hero where visitors actually test OCR scan, 100+ language AI, and weekly build metrics.
Pricing Transparency	Force users to book a sales call and wait days just to hear rough cost ranges.	**Appzys Innovation:** Real-time *4-Step App Cost & Timeline Estimator* calculating min/max budget and sprint weeks instantly, stored via REST API and attached to the lead.
Store Credibility (E-E-A-T)	Display unverified client logos or generic NDA claims without public App Store metrics.	**Appzys Innovation:** Own flagship consumer apps (~1M+ installs, 4.1★ rating, 1,500+ reviews on *Copy Text On Screen*) directly verifiable on Play Store & App Store.
IP & Code Ownership	Vague handoff clauses or junior-staffed outsourcing pyramids hidden behind account managers.	**Appzys Innovation:** Explicit *100% Source-Code IP Ownership & NDA Protection From Day 1* + weekly TestFlight / APK builds visible in the showcase & process timeline.
Service Focus	Overwhelmed menus selling 40+ unrelated IT services (SAP, ERP, SharePoint, Blockchain, Staffing).	**Appzys Innovation:** Laser-focused *Mobile App & AI/OCR Engineering Studio* with dedicated AI/OCR differentiation rather than bloated enterprise IT consulting.

EVALUATION DIMENSION	WHAT TOP-10 COMPETITORS & REFERRALS DO	WHAT IS MISSING THERE & WHAT APPZYS IMPLEMENTS
Lead Qualification Flow	Basic HTML contact forms that lose context about what features the client wanted.	**Appzys Innovation:** Clicking "Lock Estimate & Book 48-Hr Audit" forwards `?estimateId={id}` to Contact, fetches the saved estimate from MySQL, and links it atomically in one DB transaction.

2.3 Competitor Structural Patterns Retained in Appzys

PROVEN CONVERSION PATTERNS RETAINED FROM ORANGEMANTRA, MAESTRO & TOP-10 LEADERS

- **Dual Conversion Paths in Navigation:** Primary high-intent CTA (*Free 48-Hr Audit*) paired with secondary exploration anchor (*Cost Estimator*), plus sticky mobile bottom bar on viewports <768px.
- **Structured 5-Step Delivery Process:** Clear milestone transparency (*01 Discovery & 48-Hr Blueprint → 02 Interactive UI/UX Prototype → 03 Agile Weekly Sprints → 04 QA & Store Compliance → 05 Launch, ASO & Scale*).
- **Engagement Model Selection Cards:** *Fixed-Scope MVP Sprint (4–8 Weeks)*, *Dedicated Senior Pod (Monthly Retainer)*, and *AI & OCR Modernization (2–5 Weeks)* on the Services page.
- **Objection-Handling FAQ Accordion:** Pre-answers buyer hesitation on pricing, senior studio advantages vs. mega-agencies, IP ownership, and Play Store / App Store review approval.

CHAPTER 3

WEBSITE FUNCTIONAL SPECIFICATION (HOW EVERY PAGE LOOKS & WORKS)

This chapter defines the exact user interface structure, visual sections, interactive rules, and expected system behavior for every page of the Appzys web platform. Items tagged `BUILT` are already implemented and verified in Days 1–2; items tagged `PLANNED` belong to Sprint 2 (Days 3–18).

3.1 Global Navigation & Persistent Chrome `BUILT`

- **Desktop Sticky Navbar (`Navbar.jsx`):** Brand mark (Appzys logo), 4 top-level route links (`Home`, `Mobile Services`, `Case Studies`, `Contact`), smooth anchor button `Cost Estimator` (`/#estimator`), and primary action button `Free 48-Hr Audit` linking to `/contact`.
- **Responsive Mobile Drawer & Sticky Bottom Bar:** Below desktop width, links collapse into a hamburger toggle drawer, while a fixed mobile footer bar keeps `Cost Estimator` and `Free App Audit` permanently reachable with one thumb tap.
- **Global Footer (`Footer.jsx`):** Displays Studio HQ (Chennai / Kanchipuram, Tamil Nadu, India), direct email (`hello@appzys.com`), IP/NDA commitment badge, core mobile capability summary (`Android • iOS • Flutter • React Native • AI & OCR`), navigation links, and copyright.

3.2 Home Page (`/` — `HomePage.jsx`) `BUILT`

The Home Page acts as a sequential B2B conversion funnel organized into seven cohesive vertical sections:

SECTION & COMPONENT	VISUAL LAYOUT & CONTENT DISPLAYED	INTERACTIVE BEHAVIOR & PURPOSE
1. Hero Split Banner + Proof Bar	**Left:** Studio headline emphasizing mobile + AI/OCR delivery, subheadline, primary & secondary CTA buttons, and live proof strip (900K–1M+ installs, 4.1★, 100% IP ownership, Senior Team). **Right:** `InteractivePhoneShowcase.jsx` device frame.	Establishes immediate authority in the first 3 seconds without stock graphics. Both hero buttons scroll to the Cost Estimator or navigate directly to the 48-Hr Audit form.
2. Interactive Phone Showcase (3-Tab Demo) `InteractivePhoneShowcase.jsx`	Smartphone bezel housing 3 clickable mode tabs: • *OCR Screen Scan* (700K+ Installs badge, Receipt/Screen simulation, extracted fields) • *100+ Language AI* (Multi-Lang Engine, English/Spanish/Tamil output preview) • *Client Build Tracker* (Weekly Transparency metrics: `99.8%` crash-free, `< 1.2s Cold Launch`, `100% Client Git Repo`)	Clicking any tab dynamically swaps the phone screen state without page reload, displaying real latency labels (`140ms on-device` for OCR, `210ms instant sync` for Multi-Lang AI) to demonstrate live mobile performance.
3. Mobile Services Grid `appData.js`	6 structured service cards covering Native iOS/Android, Cross-Platform Flutter/RN, AI/OCR Vision, Mobile UI/UX Prototyping, App Modernization, and Store Launch & ASO.	Gives technical and non-technical buyers quick scanning of deliverables; links onward to `/services/mobile-app-development`.

SECTION & COMPONENT	VISUAL LAYOUT & CONTENT DISPLAYED	INTERACTIVE BEHAVIOR & PURPOSE
4. App Cost & Timeline Estimator `CostCalculator.jsx`	4-step interactive form: Step 1: Platform (`flutter` $5,500, `react-native` $5,500, `single-native` $4,500, `both-native` $9,000) Step 2: UI Level (`mvp` +$800/+0w, `custom` +$2,000/+1w, `design-system` +$3,500/+1w) Step 3: AI Capability (`none` $0/+0w, `ai-assistant` +$2,200/+2w, `custom-ai-ocr` +$4,000/+2w) Step 4: 6 Toggle Feature Chips (+$750 and +1w each)	Computes `minBudget`, `maxBudget = Math.round(minBudget * 1.3)`, and `estWeeks = Math.max(4, 4 + featureCount + aiWeeks + uiWeeks)` in real time. Clicking "Save Estimate & Book Audit" executes `POST /api/v1/estimates`, receives database `id`, displays the estimate reference, and links to `/contact?estimateId={id}`.
5. Mobile Tech Stack Tabs `TechStackTabs.jsx`	4 strictly mobile-focused filter tabs: 1) *Mobile Frameworks* 2) *AI & OCR Vision* 3) *Mobile UI/UX & Motion* 4) *App Store & Quality*	Replaces cluttered tech walls with focused tab switching. Strictly excludes non-mobile services (Backend/Cloud/Database) per studio positioning constraints.
6. 5-Step Delivery Process `DevelopmentProcess.jsx`	Ordered workflow cards (01 Discovery & 48-Hr Blueprint → 02 Interactive UI/UX Prototype → 03 Agile Weekly Sprints → 04 QA & Store Compliance → 05 Launch, ASO & Scale).	Reduces buyer risk by clearly explaining weekly TestFlight/APK drops, crash monitoring, and Google Play / Apple App Store approval ownership.
7. Featured Case Studies + FAQ Accordion `FaqSection.jsx`	Highlights flagship Appzys apps plus 4 expandable/collapsible Q&A rows covering cost, senior studio benefits, 100% IP ownership, and Play/App Store ASO approval.	FAQ uses smooth state toggle per question; clicking CTA buttons routes the visitor to full Case Studies or Contact.

3.3 Secondary Public Pages Already Built **BUILT**

- **Mobile App Services Page (`/services/mobile-app-development` — `MobileAppServicePage.jsx`):** Expands the 6 mobile capabilities alongside 3 selectable engagement models: *Fixed-Scope MVP Sprint (4–8 Weeks)*, *Dedicated Senior Pod (Monthly Retainer)*, and *AI & OCR Modernization (2–5 Weeks)*, each terminating in pre-contextualized audit CTAs.
- **Case Studies Showcase (`/case-studies` — `CaseStudiesPage.jsx`):** Calls `GET /api/v1/case-studies` on mount to render database-driven app case studies (*Copy Text On Screen*, *AI Text Scanner*, *Handwriting to Text Scanner*) displaying category, installs label, rating, summary, engineering challenge, implemented solution, and tech tags, with silent fallback to `FALLBACK_CASE_STUDIES` if offline.
- **Contact & Free 48-Hr Audit Page (`/contact` — `ContactPage.jsx`):** Parses URL query `?estimateId=`. When present, calls `GET /api/v1/estimates/{id}` and renders a locked Estimate Summary Banner showing platform, AI tier, budget range, and timeline. Form captures Name, Email, Phone, Company, Service Type, Budget Range, and Message plus a hidden CSS-concealed `website` honeypot field against bots. Submitting invokes `POST /api/v1/leads`, stores the lead + links `cost_estimates.lead_id`, and returns an instant booking confirmation.

3.4 Sprint 2 Planned Pages & Admin Console PLANNED

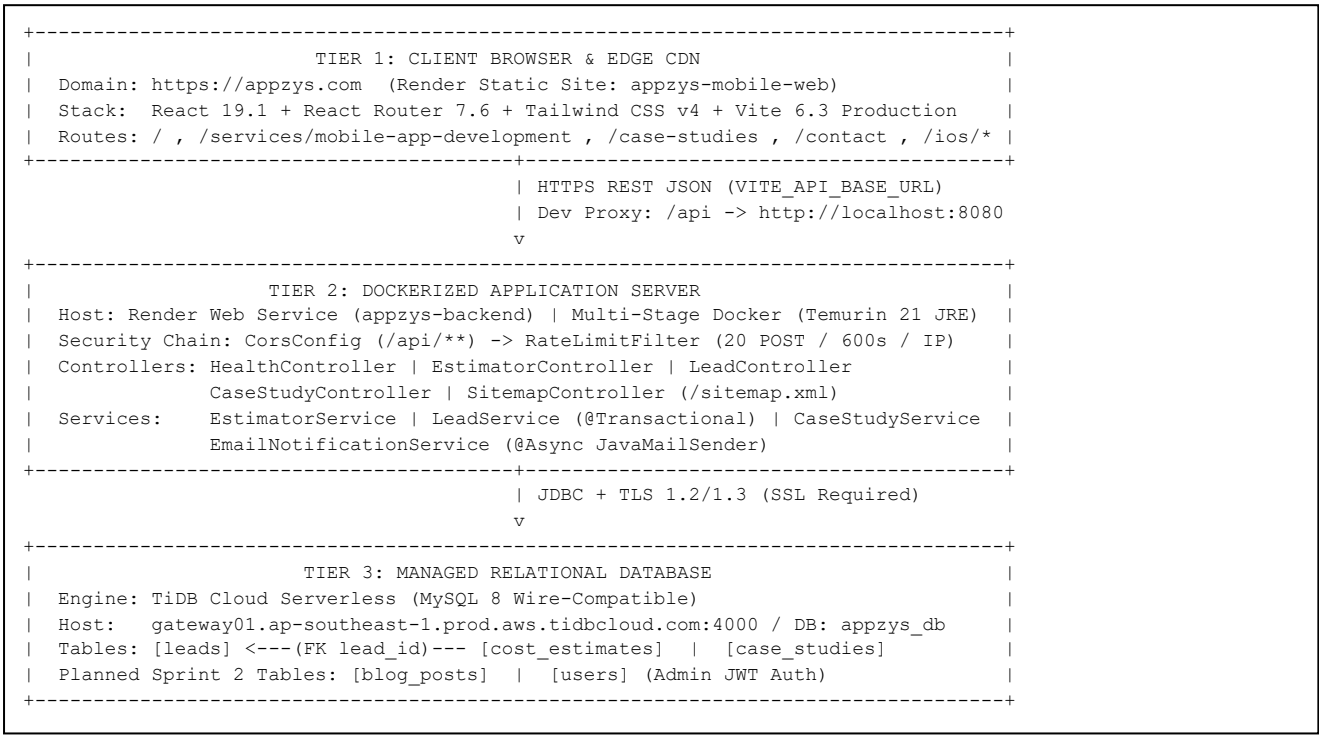
PLANNED ROUTE / MODULE	FUNCTIONAL BEHAVIOR IN SPRINT 2 (DAYS 3–18)
`/services/ai-ocr-app-development`	Dedicated flagship SEO landing page detailing on-device vs. cloud OCR pipelines, document capture SDKs, latency benchmarks, and direct AI modernization inquiry triggers.
Stack Selector Wizard	Interactive decision helper (Flutter vs. React Native vs. Native Swift/Kotlin) recommending optimal mobile architecture based on budget, animation load, and device-sensor needs.
`/case-studies/:slug`	Full dynamic Case Study detail page fetched from `GET /api/v1/case-studies/{slug}` with before/after UX metrics, Play/App Store links, and architectural diagrams.
`/about` & `/blog`	Studio story (Appzys Technologies Pvt. Ltd. mission, Chennai HQ, zero headcount mention) and Engineering Insights blog backed by `blog_posts` table for organic SEO traffic.
`/admin/**` (JWT Protected)	Internal operations portal for Appzys leadership to filter incoming leads, update status (`NEW → QUALIFIED → WON`), inspect linked estimates, and publish new case studies/blogs.
`/ios/*/privacypolicy` & `termsofuse`	Preserve all existing consumer app legal routes (`textscanner`, `handocr`, `aichat`, `copytext`) inside React Router or static routes with zero 404 regressions.

CHAPTER 4

SYSTEM ARCHITECTURE, TECH STACK & DATA FLOW DIAGRAMS

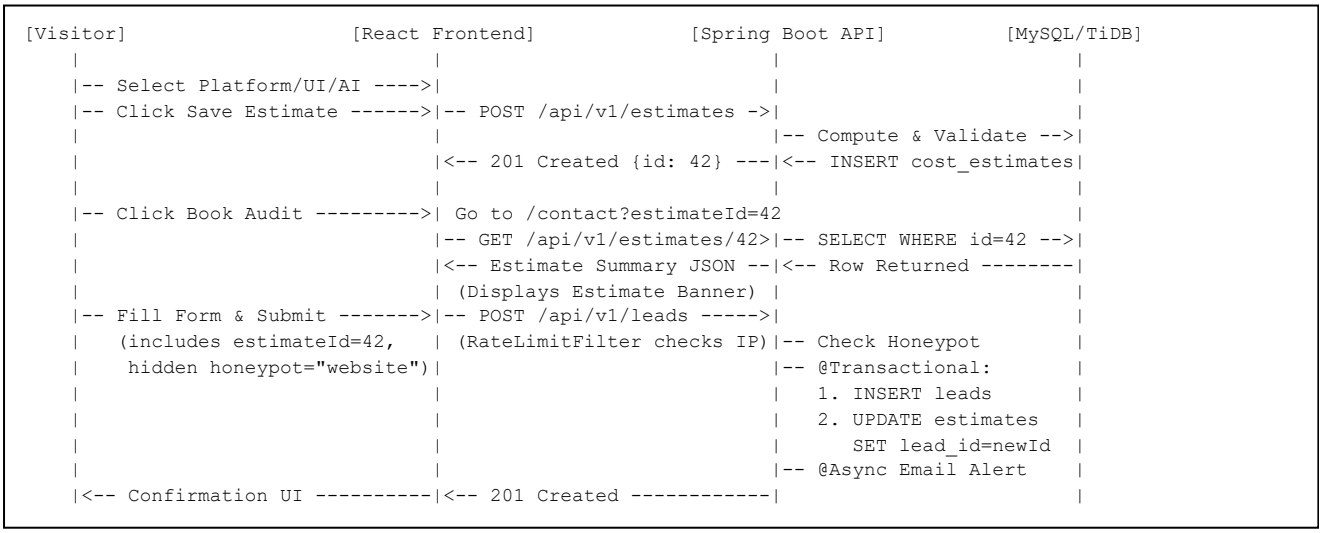
4.1 Production 3-Tier Deployment Architecture

The platform follows a decoupled 3-tier client-server architecture packaged for zero-maintenance cloud deployment on Render and TiDB Cloud Serverless. Static React assets are served over edge CDN, while business logic, validation, rate limiting, and database persistence execute inside a containerized Java 21 Spring Boot API.



4.2 End-to-End Lead & Cost Estimate Transaction Flow

A core engineering differentiator of this platform is atomic linkage between the anonymous Cost Estimator calculation and a finalized sales inquiry. Below is the exact execution sequence implemented in Days 1–2:



4.3 Complete Technology Justification Table

LAYER	SELECTED TOOL & VERSION	ENGINEERING RATIONALE
Frontend UI	React `^19.1.0`, React Router `^7.6.0`, Lucide Icons `^0.511.0`	Component-driven architecture supports interactive widgets (Phone Showcase tabs, Cost Calculator state, FAQ accordions) with clean route transitions.
Build & Styling	Vite `^6.3.5`, Tailwind CSS `^4.1.7` (`@tailwindcss/vite`)	Sub-second HMR during development and compact static bundles (`323.66 kB` JS, `36.34 kB` CSS verified via `vite build`).
Backend Framework	Java 21 LTS, Spring Boot `3.4.3` (Web, JPA, Validation, Mail)	Enterprise type safety, declarative Bean Validation on lead/estimate DTOs, Spring Data JPA repository abstraction, and native `@Async` email dispatch.
Database	MySQL 8 / TiDB Cloud Serverless (`mysql-connector-j`)	ACID transactions for relational integrity between `leads` and `cost_estimates` over TLS 1.2/1.3 without managing raw database VM patching.
DevOps & Hosting	Multi-stage `Backend/Dockerfile` + Render Static & Web	Reproducible builds using `maven:3.9.9-eclipse-temurin-21` compiled into lightweight `eclipse-temurin:21-jre-alpine` runtime image.

4.4 Exact Verified Repository Structure (Days 1–2 Baseline)

```
Mobile Application/
|-- .gitignore                            # Protects secrets & local properties
|-- Frontend/
|   |-- package.json & vite.config.js     # Port 5173 + /api -> localhost:8080 proxy
|   +-- src/
|       |-- App.jsx                       # Routes: /, /services/..., /case-studies, /contact
|       |-- data/appData.js               # 6 mobile service cards + 3 fallback case studies
|       |-- services/api.js               # REST client (VITE_API_BASE_URL || '/api/v1')
|       |-- components/                   # Navbar, Footer, InteractivePhoneShowcase,
|       |                                 # CostCalculator, TechStackTabs, Process, FaqSection
|       +-- pages/                        # Home, MobileAppService, CaseStudies, Contact
+-- Backend/
    |-- Dockerfile & pom.xml              # Spring Boot 3.4.3 + Web, JPA, Validation, Mail
    +-- src/main/
        |-- resources/application.properties # Env-var config + optional application-local
        +-- java/com/appzys/web/
            |-- AppzysWebApplication.java    # @SpringBootApplication + @EnableAsync
            |-- config/                      # CorsConfig, DataSeeder, RateLimitFilter
            |-- controller/                  # Health, Lead, Estimator, CaseStudy, Sitemap
            |-- dto/                         # Validated Request/Response + ApiException
            |-- entity/ & repository/        # Lead, CostEstimate, CaseStudy JPA + repos
            +-- service/                     # Lead, Estimator, CaseStudy, EmailNotification
```

CHAPTER 5

DATABASE SCHEMA, REST API SPECIFICATION & SECURITY CONTROLS

5.1 Relational Database Schema

The schema in `appzys_db` enforces normalized lead tracking, immutable cost audit trails, and CMS-style case study storage. Tables 1–3 are live in Phase 1; Tables 4–5 are scoped for Sprint 2.

TABLE & STATUS	COLUMNS & CONSTRAINTS	RELATIONSHIPS & PURPOSE
leads **BUILT**	`id` (PK, Auto), `name`, `email`, `phone`, `company`, `service_type`, `budget_range`, `message`, `source_page`, `status`, `created_at`	Stores all validated project/48-Hr Audit submissions. Default status tracks funnel progression inside sales pipeline.
cost_estimates **BUILT**	`id` (PK, Auto), `lead_id` (FK → `leads.id`, Nullable), `platform`, `ui_level`, `features_json`, `ai_level`, `min_budget`, `max_budget`, `est_weeks`, `created_at`	Captures anonymous calculator outputs first; atomically links `lead_id` via `@Transactional` when the visitor submits the Contact form.
case_studies **BUILT**	`id` (PK, Auto), `slug` (UNIQUE), `app_name`, `category`, `installs_label`, `rating`, `summary`, `challenge`, `solution`, `tech_stack_csv`, `featured`, `created_at`	Seeded on startup via `DataSeeder.java` with 3 real Appzys apps (*Copy Text On Screen*, *AI Text Scanner*, *Handwriting to Text Scanner*).
blog_posts **PLANNED**	`id` (PK), `slug` (UNIQUE), `title`, `excerpt`, `body_markdown`, `author`, `published`, `published_at`	Drives long-tail SEO articles (e.g., Flutter vs React Native cost, On-Device OCR latency guides) in Sprint 2.
users **PLANNED**	`id` (PK), `email` (UNIQUE), `password_hash` (BCrypt), `role`, `created_at`	Authenticates Appzys internal leadership into `/admin/**` via stateless Spring Security JWT tokens.

5.2 Complete REST API Contract

METHOD	ENDPOINT PATH	STATUS	BEHAVIOR, VALIDATION & RESPONSE CONTRACT
`GET`	`/api/v1/health`	**BUILT**	Returns service uptime payload for Render zero-downtime health checks and monitoring probes.
`POST`	`/api/v1/estimates`	**BUILT**	Server-side authoritative cost calculation from platform/UI/AI/features; persists row and returns ID, min/max budget, and weeks. Rate-limited.
`GET`	`/api/v1/estimates/{id}`	**BUILT**	Retrieves saved estimate summary so Contact page can restore quote context via `?estimateId={id}`.
`POST`	`/api/v1/leads`	**BUILT**	Validates DTO (`LeadRequest` with optional `Long estimateId`), drops honeypot bots silently, saves lead + updates `cost_estimates.lead_id`, triggers `@Async` mail alert.

METHOD	ENDPOINT PATH	STATUS	BEHAVIOR, VALIDATION & RESPONSE CONTRACT
`GET`	`/api/v1/case-studies`	**BUILT**	Returns ordered list of Appzys case studies for public Case Studies grid and featured Home blocks.
`GET`	`/api/v1/case-studies/{slug}`	**BUILT**	Returns full architectural case study record matching unique URL slug or 404 handled by `GlobalExceptionHandler`.
`GET`	`/sitemap.xml`	**BUILT**	Dynamically outputs XML sitemap covering core static routes and database-backed case study slugs for search bots.
`POST/PATCH`	`/api/v1/auth/login` & `/api/v1/admin/**`	**PLANNED**	JWT authentication + protected CRUD endpoints for lead status management, blog publishing, and case study updates.

5.3 Implemented Security & Reliability Architecture

DEFENSE-IN-DEPTH CONTROLS ACTIVE IN PRODUCTION CODEBASE

- **IP Rate Limiting (`RateLimitFilter.java`):** Enforces maximum **20 POST requests per 600 seconds (10 minutes) per client IP** on `/api/v1/leads` and `/api/v1/estimates`, inspecting `X-Forwarded-For` behind Render's reverse proxy to block automated form flooding.
- **Invisible Honeypot Spam Trap (`ContactPage.jsx` & `LeadService.java`):** Includes a hidden `website` input field untouched by real human users; automated bots populating it are silently discarded without database pollution.
- **Environment-Variable Secret Isolation:** Database URLs, TiDB credentials, and SMTP mail passwords are excluded from Git via `.gitignore` and injected via Render environment variables or gitignored `application-local.properties`.
- **Strict CORS Policy (`CorsConfig.java`):** Restricts `/api/**` cross-origin requests exclusively to whitelisted origin patterns supplied via `CORS_ALLOWED_ORIGINS`.
- **SQL Injection Prevention & Global Error Sanitization:** Spring Data JPA parameterized queries prevent SQL injection, while `GlobalExceptionHandler` returns clean JSON validation errors without leaking stack traces.

CHAPTER 6

TOP-10 SEO PLAN, QA TESTING, RISK MATRIX & 18-DAY SCHEDULE

6.1 Top-10 Search Engine Optimization (SEO) Strategy

To compete organically against multi-year mobile agencies, pure client-side React rendering alone is insufficient for fast indexation. Appzys applies a three-pillar technical SEO plan across Sprint 2:

SEO PILLAR	TECHNICAL & CONTENT EXECUTION PLAN
1. Crawlability & Pre-Rendering	Maintain live `/sitemap.xml` in Spring Boot; add Vite static HTML pre-rendering (or per-route `<title>`, canonical tag, Open Graph, and meta description injection) so Googlebot reads complete HTML immediately.
2. Structured JSON-LD Schema	Embed `Organization`, `ProfessionalService`, `SoftwareApplication` (referencing Store apps), `FAQPage` (matching `FaqSection.jsx`), and `BreadcrumbList` JSON-LD blocks.
3. "Win-First" Keyword Ladder	Target high-intent, lower-competition niches first (*"OCR mobile app development company"*, *"AI document scanner app developers"*, *"Flutter app development studio India"*) before scaling to broad terms like *"mobile app development company"* via case studies and blog authority.
4. Store Backlink E-E-A-T Loop	Cross-link Google Play and Apple App Store developer listings back to `https://appzys.com` and vice versa, giving search crawlers verifiable real-world product authority.

6.2 Quality Assurance (QA) & Verification Gates

VERIFICATION GATE	LATEST EXECUTED CHECK RESULT	SPRINT 2 AUTOMATED TEST EXPANSION
Frontend Production Build	`npm run build` (Vite 6): **PASSED** in 4.11s (`323.66 kB` JS, `36.34 kB` CSS; 0 errors).	Add Playwright / Cypress E2E smoke tests covering Estimator → Contact estimate linking and mobile responsive layout checks.
Backend Maven Compile	`mvn compile` (Java 21): **PASSED** in 3.811s (24 source files compiled; `BUILD SUCCESS`).	Add JUnit 5 + MockMvc integration tests for `RateLimitFilter` (21st POST blocked with HTTP 429), validation errors, and atomic lead linkage.
SEO & Core Web Vitals	Planned pre-launch Lighthouse audit (≥95 Performance, ≥95 SEO, 100 Accessibility).	Validate all legacy `/ios/*/privacypolicy` & `termsofuse` routes return HTTP 200 after final deploy.

6.3 Project Risk & Mitigation Matrix

IDENTIFIED RISK	SEVERITY	PREVENTIVE MITIGATION CONTROL
App Store Legal URL Breakage	Critical	Preserve exact paths (`/ios/textscanner/*`, `/ios/handocr/*`, `/ios/aichat/*`, `/ios/copytext/*`) and test HTTP 200 status before DNS cutover.
Lead Spam / API Abuse	High	Enforced dual-layer `RateLimitFilter` (20 POSTs / 10 min / IP) + hidden `website` honeypot field + DTO validation.

IDENTIFIED RISK	SEVERITY	PREVENTIVE MITIGATION CONTROL
Render Cold-Start Latency	Medium	Frontend case studies include instant `FALLBACK_CASE_STUDIES` rendering + scheduled `GET /api/v1/health` ping keeps container warm.
Service Positioning Drift	Medium	Code review checklist ensures public UI stays 100% focused on Mobile & AI/OCR development with zero internal headcount disclosure.

6.4 Master 18-Day Execution Schedule (Actual Progress + Roadmap)

TIMELINE	PHASE / MILESTONE	EXACT ENGINEERING DELIVERABLES	STATUS
Days 1–2	**Core Platform MVP & Deploy Prep**	Competitor audit, Home + Showcase + Cost Calculator + Tech Tabs + Process + FAQ + Services + Case Studies + Contact UI; Spring Boot REST API + TiDB schema + RateLimit + Async Email + Dockerfile + Render config.	**COMPLETED**
Days 3–6	**SEO Depth & Flagship AI Landing**	Launch `/services/ai-ocr-app-development`, full `/case-studies/:slug` detail view, interactive Stack Selector Wizard, legacy `/ios/*` route migration, JSON-LD + per-route meta tags.	**PLANNED**
Days 7–10	**Admin Portal & Blog CMS**	Implement Spring Security JWT auth (`users` table), `/admin` dashboard (lead status workflow & estimate inspection), and `blog_posts` API + 3 launch articles.	**PLANNED**
Days 11–14	**Automated QA & Performance**	Write JUnit/MockMvc & Playwright test suites, image/bundle compression, Google Search Console sitemap submission, Core Web Vitals optimization.	**PLANNED**
Days 15–18	**Production Cutover & Launch**	Verify SSL/DNS cutover on `appzys.com`, verify live SMTP notifications & TiDB backups, cross-link Store profiles, execute 48-hr post-launch monitoring.	**PLANNED**

CONCLUSION

With Phase 1 (Days 1–2) compiled, verified, and architected on React + Vite + Tailwind CSS and Java 21 Spring Boot + MySQL/TiDB Cloud, Appzys Technologies Private Limited possesses a production-ready foundation that solves the key conversion and transparency gaps observed across Top-10 competitors. Executing Sprint 2 (Days 3–18) completes the SEO authority layer, Admin operations console, automated test coverage, and zero-downtime domain launch of `appzys.com`.